

APPENDIX III

Physical Constants (MKSA)

c	Speed of light in vacuum	2.998×10^8 m/sec
ϵ_0	Permittivity of vacuum	8.854×10^{-12} farad/m
μ_0	Permeability of vacuum	$4\pi \times 10^{-7}$ henry/m
h	Planck's constant	6.626×10^{-34} joule . sec
k	Boltzmann's constant	1.381×10^{-23} joule/K
G	Gravitational constant	6.671×10^{-11} Nm ² /kg ²
e	Charge of proton	1.602×10^{-19} coul
m_p	Rest mass of proton	1.673×10^{-27} kg
m_e	Rest mass of electron	9.109×10^{-31} kg
m_n	Rest mass of neutron	1.675×10^{-27} kg
m_p/m_e	Proton/electron mass ratio	1.836×10^3
amu	Unified atomic mass unit	1.661×10^{-27} kg
a_0	Bohr radius	5.292×10^{-11} m
r_e	Classical electron radius	2.818×10^{-15} m
N_A	Avogadro's number	6.022×10^{23} mol ⁻¹
N_L	Loschmidt's number	2.687×10^{25} m ⁻³
V_0	Molar volume at STP	22.4×10^{-3} m ³ /mol
R	Gas constant ($N_A k$)	8.314 joule/(K mol)
g	Standard acceleration of gravity	9.807 m/sec ²